



Secrets of the chad beetle

Scientists have revealed deep secrets on the diabolical ironclad beetle's hard exoskeleton. <http://digit.in/nov20-31>



"Silent" mutations in virus strain

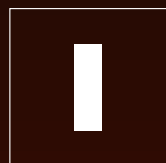
Researchers have identified a number of "silent" mutations within COVID-19's genetic code. <http://digit.in/nov20-32>



NOBEL PRIZE WINNERS 2020

This year's science prizes recognise advancements in gene editing and auction theory, as well as an improved understanding viruses, black holes

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t is that time of the year when the Nobel Prizes are given out. The announcements were made over a

few days in the second week of October. Each prize can be shared by a maximum of three people, which can be split equally between three people, or half to one person with the other two getting one fourth each. Here is a look at the 2020 winners.

PHYSIOLOGY OR MEDICINE

The prize is shared equally among

Harvey J Alter, Michael Houghton and Charles M Rice. Hepatitis is an inflammation of the liver caused by a number of factors including alcoholism, toxins and other diseases. There are also infectious forms of hepatitis, that spread from one person to another. In the 1940s, scientists had identified two forms of infectious hepatitis, the relatively harmless hepatitis A that spreads through contaminated water or food, and the more harmful hepatitis B that spreads through blood and other bodily fluids. Baruch Blumberg discovered the hepatitis B virus for which he was awarded the Nobel Prize in Physiology or Medicine in 1976. However, the disease continued to spread through blood transfusions

despite screenings for hepatitis A and B, which indicated that there was another agent that spread through the blood. Harvey J Alter demonstrated that there was an unknown virus that caused the disease to spread by conducting transfusions on Chimpanzees. Despite several attempts to isolate the virus, the scientists were not successful. Michael Houghton, used an untested method of identifying fragments of the virus in the cells of infected chimpanzees, in tandem with studies of antibodies in patients suffering from hepatitis, homed in on a new RNA virus, now called hepatitis C. This work was followed up by Charles M Rice, who proved that the hepatitis C virus alone was responsible for the spread of the disease, by cloning and replicating the virus, then introducing it to chimpanzees that then contracted the disease. The identification and isolation of the hepatitis C virus was a major advancement for medicine, paving the way for the development